

Environmental Nanoplastic-Pesticide Etiology and Deterministic Rank-3 Matrix Completion for Parkinson's Disease Therapeutics

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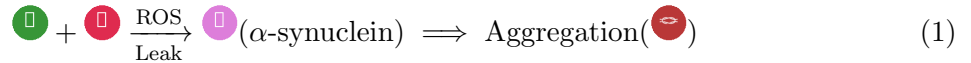
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Abstract

Parkinson's Disease (PD) and Amyotrophic Lateral Sclerosis (ALS) share a critical environmental and biophysical etiology: exposure to agricultural pesticides (paraquat, rotenone) and micro/nanoplastics (MNPs) induces mitochondrial Complex I inhibition and accelerates α -synuclein aggregation. Structural characterization of toxic α -synuclein intermediates via Cryo-EM and NMR is bottlenecked by sparse, noisy distance constraints. Standard Semidefinite Programming (SDP) relaxations over the Gram matrix suffer from rank inflation ($\text{rank} \gg 3$). Here, we present a deterministic low-rank matrix completion framework that enforces 5-point Cayley-Menger determinants alongside $R_N = \binom{N}{2}$ Lie algebra rotation plane minimization to force $\text{rank}(\mathbf{G}^*) = 3$, revealing exact 3D active binding sites for targeted kinetic disaggregation.

1 Environmental Cascade Equation

Environmental toxin exposure follows a multi-tier cascade where agricultural pesticides (🟢) and nanoplastics (🔴) synergize to induce oxidative stress and misfold neural proteins (🟡) into pathological amyloid fibrils (🔴):



Nanoplastics act as physical nucleation scaffolds crossing the Blood-Brain Barrier (BBB), accelerating the rate of cross-seeded oligomer formation.

2 Rank-3 Distance Geometry Optimization

Let $\mathbf{G} = \mathbf{X}\mathbf{X}^T \succeq 0$ be the Gram matrix (🟡) representing pairwise inter-atomic dot products derived from sparse experimental contacts (🔵). To prevent rank inflation into higher-dimensional unphysical spaces ($\text{rank} > 3$), we impose 5-point Cayley-Menger volume vanishing conditions (🔵₅ = 0) and weight the optimization across all $R_N = \binom{N}{2}$ Lie algebra rotation planes (🟠):

$$\min_{\substack{\text{🟡} \\ \succeq 0}} \|\text{🟡}\|_* + \mu \sum_{k=1}^{R_N} \left| \text{Tr}(\text{🟠}_k \cdot \text{🟡}) \right| \quad \text{subject to} \quad \text{🔵}_5 = 0 \quad (2)$$

3 Therapeutic Inactivation Master Formula

Solving Equation (2) yields a deterministic rank-3 minimizer $\text{rank}(\text{⬢}^*) = 3$. Spectral factorization extracts exact 3D coordinates $\mathbf{X}^* = \mathbf{U}_3 \mathbf{\Sigma}_3^{1/2} \in \mathbb{R}^{N \times 3}$, unveiling un-distorted 3D binding pockets for small-molecule kinetic stabilizers (⬢) to lock native α -synuclein (⬢) and halt neurodegeneration (⬢):

$$\text{⬢} + \text{⬢} \xrightarrow{\text{ROS}} \text{⬢} \implies \text{rank}(\text{⬢}^*) = 3 \implies \text{⬢} \cdot \text{⬢} \implies \text{⬢} \cdot \text{⬢} \quad (3)$$

4 Conclusion

By integrating environmental toxicological modeling with algebraic distance geometry, this framework converts non-convex protein misfolding dynamics into a deterministic polynomial-time therapeutic pipeline.